

Governing Equations & Nondimensionalization

1. Dimensional Governing Equations

Heat Equation

The heat conduction equation for constant thermal conductivity applies across the spatial domain $r \in [R, R_s]$:

$$\frac{\partial T}{\partial t} = \alpha \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = \alpha \left(\frac{1}{r} \frac{\partial T}{\partial r} + \frac{\partial^2 T}{\partial r^2} \right) \quad (1)$$

where $\alpha = \frac{\Gamma}{\rho_s C_s}$ is the thermal diffusivity of the soil.

Pore-Pressure Equation

Combining soil grain and water mass balance formulations yields the governing second-order parabolic PDE for excess pore pressure u in the soil mass:

$$-[(1-n)\beta_p + n\beta_w] \frac{\partial T}{\partial t} + \left(n\alpha_w + \frac{1}{K_s} \right) \frac{\partial u}{\partial t} - \frac{k}{\gamma_w} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) = 0 \quad (2)$$

where β_p is the thermal expansion coefficient of the solid particles, β_w is the thermal expansion coefficient of water, $\gamma_w = \rho_w \cdot g \approx 9810 \text{ Pa/m}$, k is hydraulic permeability, α_w is water compressibility, and K_s is the bulk modulus of soil grains.

Dimensional Initial and Boundary Conditions

- **Pore Pressure Conditions:**

$$\left(\frac{\partial u}{\partial r} \right)_{r=R} = 0, \quad u(r = R_s, t) = 0 \quad \text{for all } t \geq 0 \quad (3)$$

$$u(r, t = 0) = 0 \quad \text{throughout the domain} \quad (4)$$

- **Temperature Conditions:**

$$T(r = R, t) = T_f, \quad T(r = R_s, t) = T_s \quad \text{for all } t \geq 0 \quad (5)$$

$$T(r, t = 0) = T_s \quad \text{throughout the domain}$$

2. Nondimensional Variables & Material Groups

Define the dimensionless space (ξ), time (τ), temperature (Θ), and pore pressure (U):

$$\xi = \frac{r}{R_s}, \quad \tau = \frac{t}{t_E}, \quad \Theta = \frac{T - T_s}{T_f - T_s}, \quad U = \frac{u}{u_{\text{ref}}}$$

where t_E is the fixed total simulation timeframe (15 days = 1.296×10^6 s).

Shorthand composite parameters:

$$\bar{\beta} = (1 - n)\beta_p + n\beta_w \quad [1/^\circ\text{C}], \quad S = n\alpha_w + \frac{1}{K_s} \quad [1/\text{Pa}]$$

3. Dimensionless Scaling Parameters

$$\alpha_n = \frac{\alpha t_E}{R_s^2} \quad (\text{Fourier number — temporal scale factor for heat diffusion})$$

$$D = \frac{k}{\gamma_w \alpha S} \quad (\text{Diffusivity ratio — hydraulic vs. thermal dissipation speed})$$

$$u_{\text{ref}} = \frac{\bar{\beta}(T_f - T_s)}{S} \quad (\text{Undrained pressure scale: theoretical max pressure when } k -$$

4. Derivation of Non-Dimensional PDEs

A. Non-Dimensional Heat Equation

Substituting ξ , τ , Θ into Eq. (1) and dividing by $\frac{T_f - T_s}{t_E}$:

$$\boxed{\frac{\partial \Theta}{\partial \tau} = \alpha_n \left(\frac{1}{\xi} \frac{\partial \Theta}{\partial \xi} + \frac{\partial^2 \Theta}{\partial \xi^2} \right)}$$

B. Non-Dimensional Excess Pore-Pressure Equation

Substituting ξ , τ , Θ , U into Eq. (2) and multiplying through by $\frac{t_E}{S \cdot u_{\text{ref}}}$:

$$-\underbrace{\frac{\bar{\beta}(T_f - T_s)}{S \cdot u_{\text{ref}}}}_{=1} \frac{\partial \Theta}{\partial \tau} + \frac{\partial U}{\partial \tau} - \underbrace{\frac{k \cdot t_E}{\gamma_w R_s^2 S}}_{=D \cdot \alpha_n} \frac{1}{\xi} \frac{\partial}{\partial \xi} \left(\xi \frac{\partial U}{\partial \xi} \right) = 0$$

Yielding the final operational PINN equation:

$$\boxed{\frac{\partial U}{\partial \tau} = D \cdot \alpha_n \left(\frac{1}{\xi} \frac{\partial U}{\partial \xi} + \frac{\partial^2 U}{\partial \xi^2} \right) + \frac{\partial \Theta}{\partial \tau}}$$

5. Non-Dimensional Initial and Boundary Conditions

$$\begin{aligned} \Theta \left(\xi = \frac{R}{R_s}, \tau \right) &= 1, & \Theta(\xi = 1, \tau) &= 0, & \Theta(\xi, \tau = 0) &= 0 \\ \left(\frac{\partial U}{\partial \xi} \right)_{\xi=R/R_s} &= 0, & U(\xi = 1, \tau) &= 0, & U(\xi, \tau = 0) &= 0 \end{aligned}$$